EXPERIMENT 3

Design of Test Cases and Maintenance Plan for a Sample Application

1. Objective

To design comprehensive test cases and prepare a maintenance plan for the Smart Library Management System to ensure software quality, reliability, maintainability, and long-term operational support.

2. Introduction

Testing is an essential activity in the Software Development Life Cycle (SDLC). It verifies that software meets functional and non-functional requirements before deployment. A well-designed test plan identifies defects early, improves software quality, and increases user confidence. After deployment, software maintenance ensures the application continues to operate efficiently by correcting defects, adapting to environmental changes, improving performance, and preventing future issues. The Smart Library Management System is used as the sample application for demonstrating testing and maintenance activities.

3. Test Cases

ID	Module	Scenario	Input	Expected Result	Status
TC001	Login	Valid Login	Valid credentials	Dashboard opens	Pass
TC002	Login	Invalid Login	Wrong password	Error displayed	Pass
TC003	Registration	Add Member	Valid details	Member added	Pass
TC004	Books	Add Book	Valid ISBN	Book stored	Pass
TC005	Search	Search Book	Book title	Book listed	Pass
TC006	Issue	Issue Book	Member & Book ID	Issue successful	Pass

TC007	Return	Return Book	Book ID	Return updated	Pass
TC008	Fine	Late Return	5 days late	Fine generated	Pass
TC009	Reports	Generate Report	Date range	PDF/Report generated	Pass
TC010	Logout	Logout	User session	Redirect to login	Pass

4. Maintenance Plan

The maintenance plan defines activities performed after software deployment to ensure continuous availability, reliability, security, and performance.

Corrective Maintenance: Fix defects reported by users after deployment.

Adaptive Maintenance: Modify the software to support changes in hardware, operating systems, browsers, or databases.

Perfective Maintenance: Enhance existing features and improve performance based on user feedback.

Preventive Maintenance: Refactor code, update libraries, improve documentation, and strengthen security to reduce future failures.

5. Maintenance Schedule

Activity	Frequency	Responsible Team
Database Backup	Daily	DB Administrator
Performance Monitoring	Weekly	DevOps
Security Updates	Monthly	System Administrator
Bug Review	Bi-weekly	Developers
Regression Testing	After every update	QA Team
Documentation Update	As Required	Documentation Team

6. Roles and Responsibilities

- Project Manager: Plans maintenance activities and monitors progress.
- Software Developer: Fixes defects and develops enhancements.
- QA Engineer: Performs functional and regression testing.
- Database Administrator: Maintains database integrity and backups.
- System Administrator: Deploys updates and monitors server health.
- Technical Support Engineer: Resolves user issues and records incidents.

7. Risk Management

Risk	Impact	Mitigation	Owner
Requirement Changes	High	Review requirements regularly	PM
Software Bugs	High	Continuous testing	QA
Data Loss	High	Daily backup	DBA
Schedule Delay	Medium	Weekly monitoring	PM
Security Threats	High	Patch management	Sys Admin

8. Conclusion

The designed test cases validate the core functionalities of the Smart Library Management System and ensure that the software satisfies specified requirements. The maintenance plan provides a structured approach for correcting defects, adapting to technological changes, improving performance, and ensuring long-term software reliability. Proper testing and maintenance contribute significantly to software quality, user satisfaction, and system sustainability.